MidiPad User Manual

version 0.21

1. Welcome

MidiPad turns a game controller into a MIDI instrument and, above all, a MIDI looper: sticks, triggers and the touchpad send continuous controllers (CC), buttons play notes and chords, and every channel can carry its own layout. It runs as a standalone program next to your DAW or synthesizer and sends MIDI to any port on your system.

MidiPad was designed around the Sony DualSense (PlayStation 5) controller, whose touchpad tracks two fingers. Other controllers recognised by SDL3 work too, without the touchpad features.

A MIDI looper in your hands

Looping is what MidiPad does best. Hold one button, play, release: what you just played repeats at once, in time with itself, and your hands are free again to play over it. No DAW recording, no clip to set up, no bar count to choose beforehand.

- **Loop notes and chords:** a chord progression, a bass line or a riff played on the buttons, the D-pad or the touchpad pads loops with its exact timing and velocity.
- **Loop movements:** a stick gesture, a trigger squeeze or a finger slide on the touchpad loops as a CC curve, a hand-drawn LFO for a filter, a pan or an effect send.
- **Layer a whole arrangement:** each of the sixteen MIDI channels has its own loops, and all of them play together. Loop chords on channel 1, switch to channel 2 and loop a bass line, then add a filter sweep on top.
- **Keep playing live:** movement loops play on CCs of their own, beside the control that recorded them, so the stick or finger stays free to play live over its own loop.
- **Shape it as you go:** add notes with overdub, play loops forwards, backwards or back and forth, pause and restart every loop at once with Play and Stop, and stretch a movement loop up to 100 times longer or 20 times shorter, or narrow its swing, while it plays.

Because the loops are plain MIDI, they drive any instrument: a plugin in your DAW, a standalone synthesizer or a hardware synth on a MIDI interface. Chapter 9 explains looping in detail.

What you can do with MidiPad

- **Play notes and chords** from the face buttons, the D-pad, the shoulder buttons and the four zones of the touchpad, momentarily or latched (toggle).
- **Sculpt sounds** with the two sticks, the analog triggers L2/R2 and two touchpad fingers, each sending a CC with its own range and direction.
- **Freeze** a stick at its current value, so a filter or effect stays put while your hands do something else.
- **Record loops** of notes and movements on every channel, and layer them live (see above).
- **Play in time** with your DAW through Ableton Link: loops snap to whole beats or bars and follow the tempo. Or leave Link off and play free.
- **Switch MIDI channels** on the fly, each channel with its own mapping, so one controller drives a whole multitimbral setup.
- **Change programs** on your synthesizer from a button or by sweeping a stick through a range of presets.
- **Humanize** velocity and timing for a less mechanical feel.

Who this manual is for

This manual assumes you know what MIDI notes, channels and CCs are, and that you have a DAW or synthesizer on your computer. It does not assume you have used MidiPad before: chapters 2 and 3 take you from installation to your first sound, and the later chapters describe every part of the window in order.

The manual describes MidiPad on **Linux**. Notes for Windows and macOS will appear in boxes marked *Windows* and *macOS* when those versions are available.

Conventions

Notation	Meaning
Options	A label, button or menu entry as it appears on screen
☰ › **Options**	A menu path: open the ☰ menu, then choose Options
`Ctrl+S`	A keyboard shortcut

Notation	Meaning
Cross, Circle, Square, Triangle	The DualSense face buttons (A, B, X, Y on other controllers)
L1/R1, L2/R2, L3/R3	Shoulder buttons, analog triggers, stick clicks
CC 40	MIDI Control Change number 40
C4 = 60	Note names: middle C is C4, MIDI note 60

> **Note:** boxes like this one give tips and details you can skip on a first reading.

2. Installation on Linux

MidiPad is built from source with CMake; the build downloads its own libraries, so you only need a compiler and the ALSA headers. Installation takes about five minutes on a recent machine.

Requirements

Item	Version or package
Compiler	C++20: GCC 13 or newer (tested), or a recent Clang
Build tools	CMake 3.25 or newer, Ninja, Git
ALSA headers	`libasound2-dev` (Debian, Ubuntu), `alsa-lib-devel` (Fedora)
JACK	Optional
Controller	DualSense over USB or Bluetooth; any SDL3 gamepad without touchpad

On Debian or Ubuntu, install everything in one command:

```
sudo apt install build-essential cmake ninja-build git libasound2-dev
```

Building

1. Get the source code and enter its folder.
2. Configure, build and (optionally) run the tests:

```
cmake --preset release
cmake --build --preset release
ctest --preset release
```

The program is created at `build/release/src/app/midipad`. You can start it from there, or copy it to a folder on your `PATH`, such as `~/.local/bin`.

Giving MidiPad access to the touchpad

The sticks, triggers and buttons work as soon as the controller is connected. The DualSense touchpad is read directly from the device (`/dev/hidraw*`), which needs a udev rule. Without it, MidiPad works but the TOUCHPAD module stays inactive, and MidiPad tells you so when the controller connects.

Create the file `/etc/udev/rules.d/70-dualsense.rules` with this content:

```
# USB
KERNEL=="hidraw*", SUBSYSTEM=="hidraw", ATTRS{idVendor}=="054c",
ATTRS{idProduct}=="0ce6", MODE="0666"
# Bluetooth
KERNEL=="hidraw*", SUBSYSTEM=="hidraw", KERNELS=="0005:054C:0CE6.*",
MODE="0666"
```

Then reload the rules and reconnect the controller:

```
sudo udevadm control --reload-rules && sudo udevadm trigger
```

Connecting the controller

- **USB:** plug in the cable. The controller is ready at once.
- **Bluetooth:** hold **Create** and the **PS** button until the light bar flashes, then pair *DualSense Wireless Controller* in your system's Bluetooth settings.

MidiPad finds the controller whether you connect it before or after starting the program. The status bar at the bottom right of the window shows its name and a green dot when it is connected.

First launch

Start MidiPad:

```
build/release/src/app/midipad
```

On the very first launch, MidiPad does not know where to send its MIDI yet, so the **MIDI output** window asks you to choose a port. Chapter 3 explains which one to pick. Until you choose, MidiPad sends to its own virtual port, named *MidiPad*.

> **Note:** MidiPad keeps reading the controller while its window is in the background, so you can play while your DAW has the focus.

> **Windows / macOS:** installation notes will be added with those versions.

3. Quick start

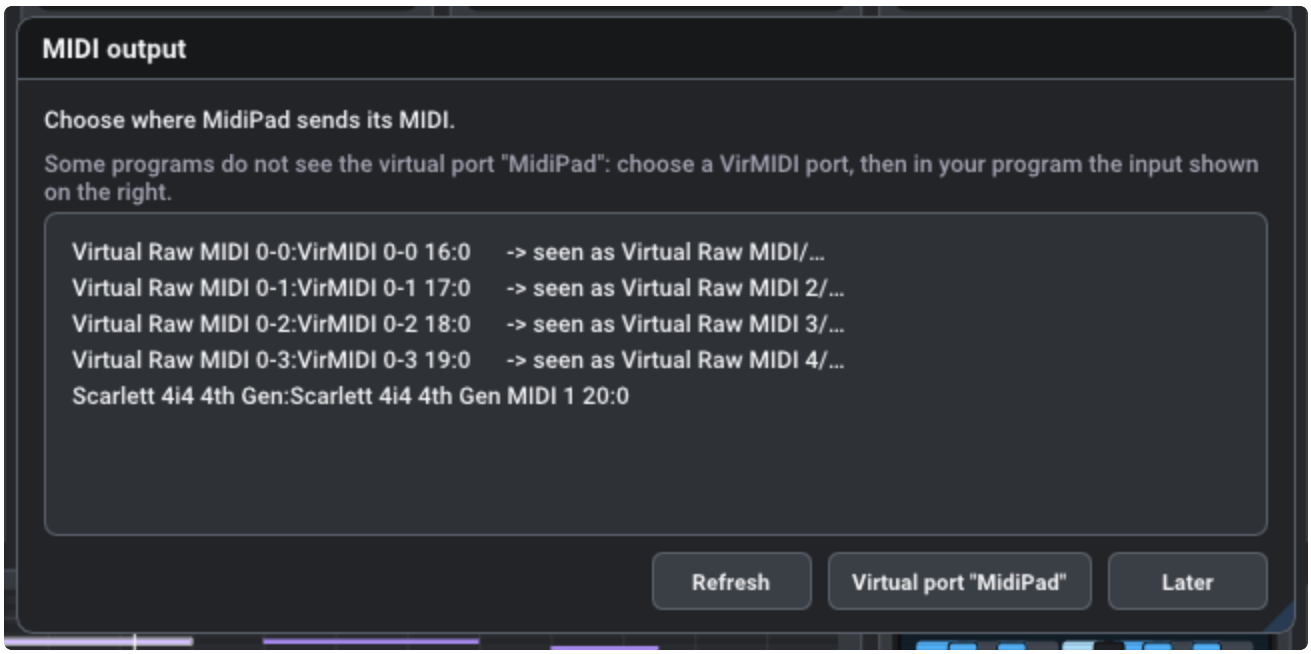
Five steps take you from a fresh installation to playing a synthesizer in your DAW: connect, choose an output, set up a track, play, save.

Step 1: connect the controller and start MidiPad

Connect the DualSense by USB or Bluetooth, then start MidiPad. The green dot at the bottom right confirms that the controller is seen.

Step 2: choose where the MIDI goes

On the first launch, the **MIDI output** window lists every MIDI port on your system.



Pick the port your DAW can read:

- **Virtual port "MidiPad"**: a software port. DAWs that use the ALSA sequencer or JACK (Reaper, Ardour, Qtractor) see it directly as an input named *MidiPad*.
- **VirMIDI ports**: kernel MIDI ports, seen by every program, including those that only list hardware devices. Choose **VirMIDI 0-0**, then select the input shown on the right (*Virtual Raw MIDI*) in your DAW. If no VirMIDI port exists, the window offers to create them (see chapter 11).
- **Later**: keep the virtual port for now; the question comes back at the next start.

MidiPad remembers your choice and reconnects to the same port at every start. You can change it at any time in ☰ › **Options**.

> **Tip:** if you are unsure, start with VirMIDI 0-0. It works with every DAW on Linux.

Step 3: set up a track in your DAW

1. Create an instrument track with a synthesizer.
2. Set the track's MIDI input to the port chosen in step 2, on all channels or channel 1.
3. Arm or monitor the track so it plays what it receives.

Step 4: play

- Press **Cross**: the chord shown under the Cross pad sounds. The pad lights up while the chord plays.

- Move the **left stick**: it sends CC 40 (left-right) and CC 41 (up-down). Use your DAW's MIDI learn to assign them to a filter cutoff or any parameter.
- Press a **D-pad** direction: in the demo preset, Up, Right, Down-left and Left latch chords (press again to stop them).

Every change you hear is also shown in the window: the stick's screen follows your thumb, the CC pills light up, and the piano roll scrolls the notes you play.

Step 5: make it yours and save

Click any control in the window to change what it sends: a pad, a D-pad row, a CC pill, a module's screen. Chapter 6 describes these mapping windows. When you are happy, press `Ctrl+S` or click **Save preset**.

> **Note:** changes are heard at once but written to disk only when you save. Unsaved changes are kept if you only switch channels; MidiPad asks before discarding them when you open another preset or quit.

4. The main window

MidiPad has a single window that plays and edits at the same time: every module mirrors a part of the controller, shows what it sends, and opens its settings when you click it. The window is built for the stage, so everything you need while playing is one click away.

The screenshots in this manual use a preset called *Electronica Demo*, in F minor, with loops playing.



#	Area	What it does
1	☰ menu	Options, Fullscreen (`F11`), Quit (`Ctrl+Q`)
2	Channel selector	The active MIDI channel (1–16); arrows step through channels
3	Preset menu	Opens, creates, saves and deletes presets; arrows load the previous or next preset
4	Play / Stop	Restarts or pauses the loops of every channel; Play turns green while loops run. With Ableton Link, they can also start and stop the Link session

#	Area	What it does
5	LINK	FREE while MidiPad keeps its own time; with Ableton Link on, the session's tempo, the number of peers and the beats of the bar. Click to turn Link on or off
6	PANIC	Silences everything at once: all notes off, loops stopped, delayed notes dropped
7	LEFT STICK	Stick position, its CCs, the stick-click (L3) action and its loop
8	RIGHT STICK	The same for the right stick (R3)
9	TRIGGERS	L2 and R2 travel, their CCs, their button actions and their loop
10	D-PAD	The eight directions, including diagonals, and what each plays
11	TOUCHPAD	Finger positions and their CCs, or the four pad zones, and the finger loops
12	Piano roll	The notes of the active channel, scrolling or as a loop clip
13	SCALE	The tonality; click to choose a root and a scale
14	CHANNEL	The active channel in large digits and the sixteen channels; a violet dot marks channels with loops
15	Button strip	One pad per button: L1, L2, L3, Create, Cross, Circle, Square, Triangle, Options, PS, Touchpad click, Mute, R3, R2, R1
16	MIDI out	The port MidiPad sends to, and the name your DAW sees it under
17	Controller	The connected controller, and *+ touchpad* when the touchpad is readable

Reading the window

- **A green dot** in a module's header means the controller is connected and that part is active.

- **Pills** such as *X CC 40* show what an axis sends; they light up as the value moves. *inv* marks an inverted axis, *PC 1-16* an axis that selects programs.
- **Pads** in the button strip flash on every press and stay lit while their note or chord sounds. A latched (toggle) chord stays lit until released. The label under each pad shows its action: a chord name, a note, *Freeze*, *Ch +1*, *Rec*, *PC 1* and so on; - means nothing is assigned.
- **The ↻ icon** next to a D-pad chord marks a toggle: press once to start, again to stop.

Clicking to edit

Click a control to open its mapping window over the view: a pad, a D-pad row, a CC pill, a module's screen, a LOOP panel or the piano roll. The window edits the mapping of the active channel if it has one, otherwise the base mapping shared by all channels. The view keeps playing behind it, so you hear each change at once.

> **Note:** the controller drives MIDI only. You operate the window with the mouse and keyboard; the controller never moves through the interface, so playing never changes a setting by accident.

5. Control modules

Each module stands for one part of the controller and has the same layout from top to bottom: an animated screen that follows your hands, a MIDI area showing what is sent, and a LOOP panel for the recorded movements.

Sticks

Each stick sends two CCs, one per axis. In the demo preset the left stick sends CC 40 (X) and CC 41 (Y), the right stick CC 42 and CC 43, with Y inverted so that pushing up raises the value. A new preset starts empty: every axis and button is unassigned until you set it.

- **Screen:** a glowing shape follows the stick. Its size grows with the distance from the centre.
- **Pills:** *X CC 40* and *Y CC 41 inv* show the CC numbers; *inv* marks an inverted axis.
- **Stick click row:** shows the action of L3 or R3, with *free* or *frozen* on the right.

A stick sends a value only when it changes, so a stick at rest adds no traffic.

Freeze

A button set to **Axis freeze** (L3 and R3 in the demo) holds the chosen axes at their current value: one press freezes, the next one releases. While frozen, the shape turns ice blue and keeps following your thumb, a ghost and a crosshair mark the frozen position, and the screen shows **FROZEN**. Bring the stick back to the crosshair before releasing if you want the value to continue smoothly.

Triggers

L2 and R2 are both analog axes and buttons. As axes they send a CC over their whole travel (CC 50 and CC 51 in the demo). As buttons they can play a note, a chord or any other action: they press at 25 % of the travel and release at 15 %, so a light touch does not trigger them by accident.

The TRIGGERS module shows both travels side by side, the CC pill and the button action of each trigger, and a LOOP panel with both recorded curves.

D-pad

The D-pad plays eight directions: the four arrows and the four diagonals. Each row of the module shows a direction, its action and, for a toggle, the ↻ icon. The screen shows a 3D D-pad with the pressed direction raised in red, and the name of what it plays.

Your thumb never presses or releases the two arrows of a diagonal at exactly the same moment. To keep the arrow crossed on the way silent, a new direction plays only after it has stayed stable for a short time: 15 ms by default, adjustable in Options. Releasing the whole D-pad stops the sound at once.

Touchpad

The DualSense touchpad tracks two fingers. Each finger works in one of three modes, chosen in the touchpad window:

Mode	What the finger does
Off	Nothing
X/Y position → CC	Sends two CCs: left-right and bottom-top (finger 1: CC 60/61, finger 2: CC 62/63 in the demo)

Mode	What the finger does
4 zones → pads	Splits the touchpad into four quarters; touching a quarter plays its note or chord

The screen shows each finger as a numbered dot. In pad mode it shows the four zones and lights the one being touched.

> **Note:** the touchpad needs the udev rule described in chapter 2. Without it, the module stays inactive and the status bar does not show *+ touchpad*.

Button strip

The strip at the bottom has one pad per button, in this order: L1, L2, L3, Create, Cross, Circle, Square, Triangle, Options, PS, Touchpad click, Mute, R3, R2, R1. The face buttons use their PlayStation colours. Click a pad to choose its action.

Piano roll, SCALE and CHANNEL

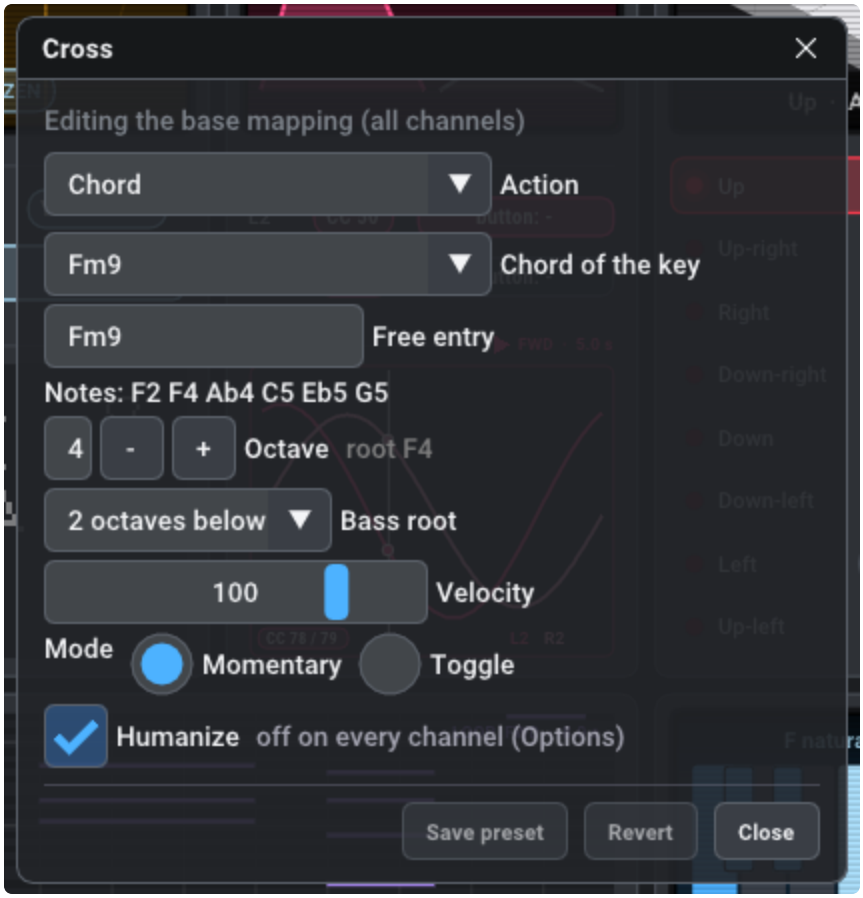
Below the modules, three blocks show the musical state of the active channel:

- **Piano roll:** the notes and chords played, scrolling towards the *now* line on the right; when a note loop runs, it turns into a clip of the whole loop (chapter 9).
- **SCALE:** the chosen tonality on a one-octave keyboard, with the notes of the scale lit. *no scale* means none is chosen. Click it to choose one (chapter 7).
- **CHANNEL:** the active channel in large digits, and a grid of the sixteen channels. A violet dot marks every channel where a loop is playing.

6. Mapping windows

A mapping window sets what one control sends. Click a control in the main window to open it; the window floats over the view, which keeps playing, so you hear every change as you make it.

Anatomy of a mapping window



- **Title:** the control being edited (here *Cross*).
- **Scope line:** *Editing the base mapping (all channels)*, or *Editing the mapping of channel N* when the active channel has its own mapping (chapter 8).
- **Action:** the type of action; the fields below change with it.
- **Save preset** (`Ctrl+S`) writes the preset to disk. **Revert** reloads the saved version and discards your changes. **Close** (or `Esc`) closes the window and keeps your changes in memory.

Button actions

Every button, D-pad direction, trigger (as a button) and touchpad zone can take one of these actions:

Action	Sends or does	Main settings
No action	Nothing	—
Note	One note	Note, Velocity, Mode, Humanize

Action	Sends or does	Main settings
CC	A Control Change	CC number, Value on press, Value on release, Mode
Chord	Several notes at once	Chord, Octave, Bass root, Velocity, Mode, Humanize
Transport	MIDI Start, Stop or Continue	Command
Channel change	Moves to the previous or next channel	Previous (-1) or Next (+1)
Axis freeze	Holds axes at their current value	Axes to freeze
Loop record	Records loops while held	— (see chapter 9)
Program change	Selects a program on the synthesizer	Program, Next or Previous, Bank select (chapter 10)
Loop stretch	While held, sticks and fingers stretch or narrow their loops	— (see chapter 9)

Mode: Momentary or Toggle

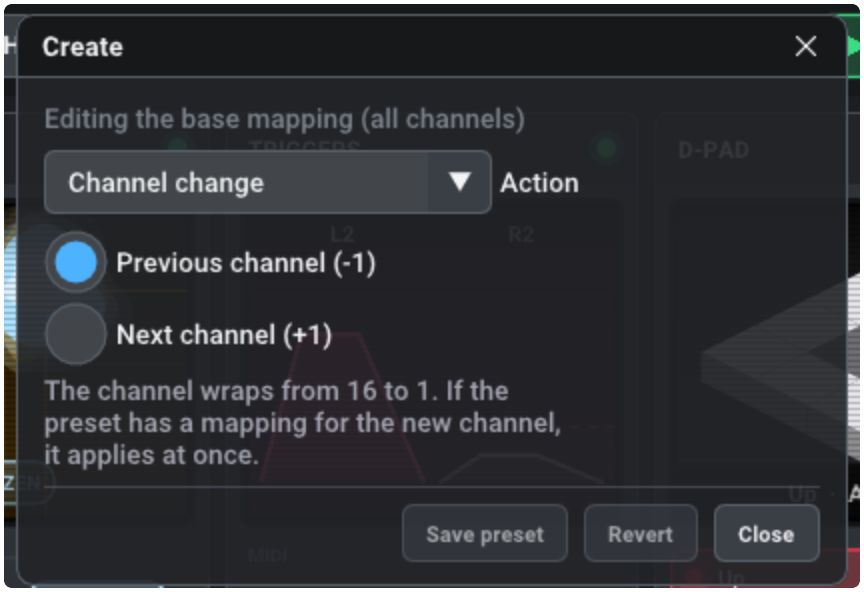
- **Momentary:** the note, chord or CC lasts as long as you hold the control. A CC sends *Value on press*, then *Value on release*.
- **Toggle:** the first press starts, the second stops. A latched note or chord keeps sounding with your hands free.

Toggles within one family of controls exclude each other: starting a latched chord on the D-pad stops the one already latched there. The families are the buttons, the D-pad, and each touchpad finger. CC toggles are always independent.

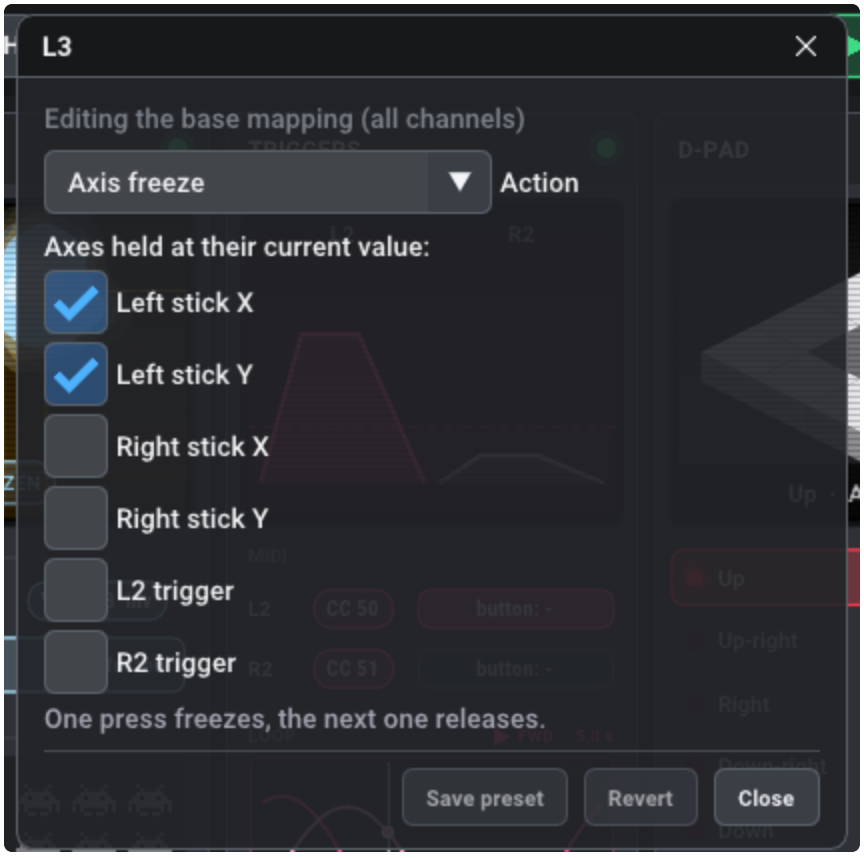
Velocity and Humanize

Velocity (1–127) sets how hard the note is played. The **Humanize** box lets this control follow the random velocity and timing set for its channel in Options (chapter 11); the text beside it tells you whether humanization is active on any channel. Untick it to keep this control exact, for a kick drum for example.

Other button windows



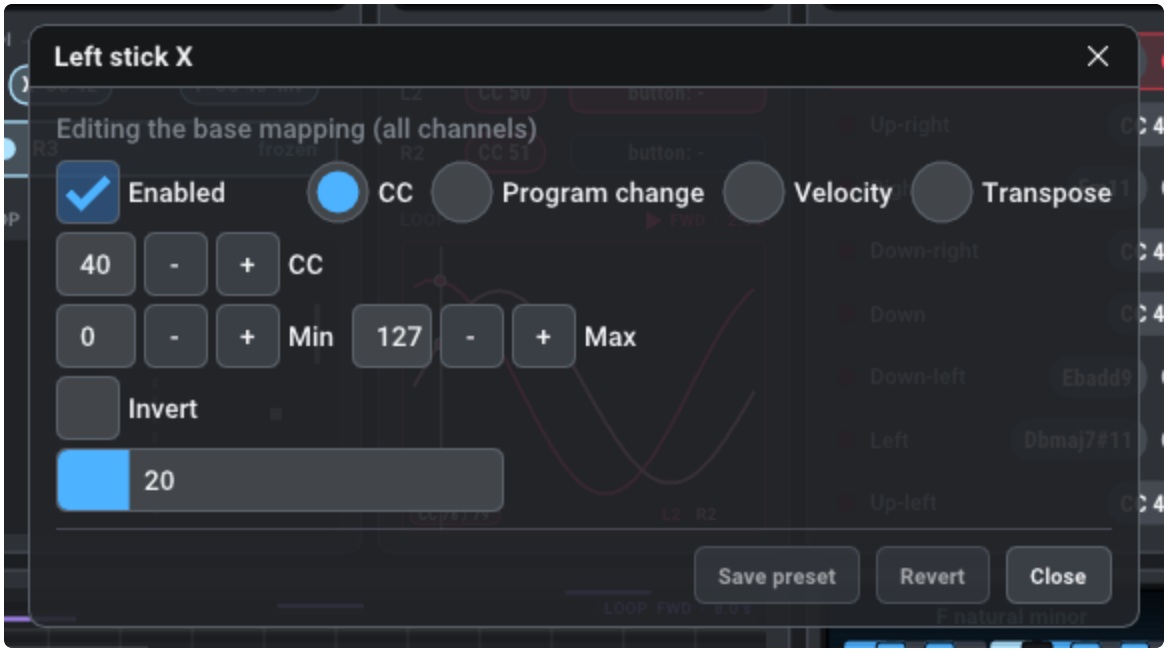
Channel change steps through the sixteen channels, wrapping from 16 to 1. If the preset has a mapping for the new channel, it applies at once.



Axis freeze holds any combination of the six axes: left stick X/Y, right stick X/Y, L2 and R2. One press freezes, the next releases.

Axis windows

Click a stick's screen, the TRIGGERS screen or the TOUCHPAD screen to edit its axes together. Click a single pill (such as *X CC 40*) to edit that axis alone.



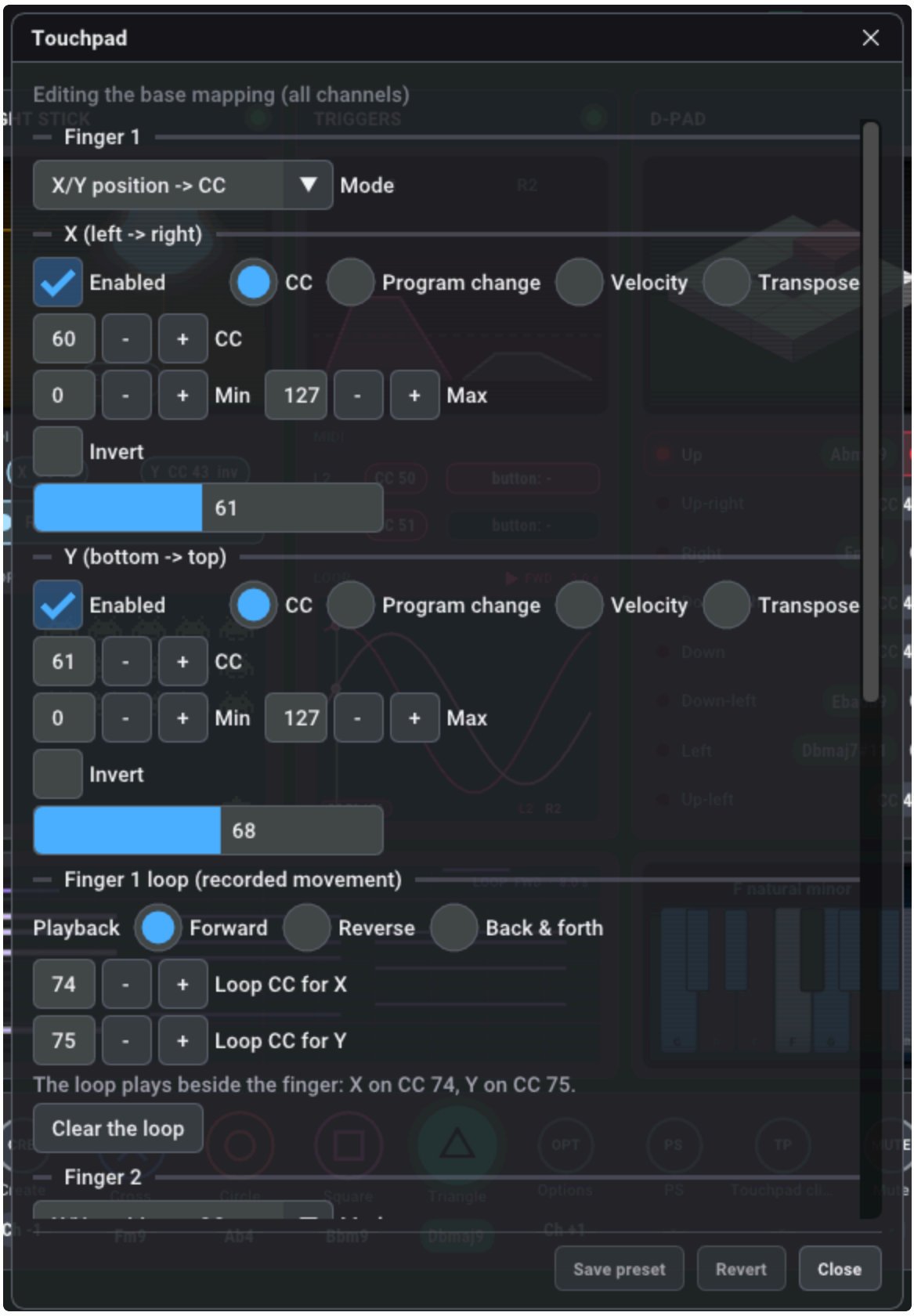
Each axis has these settings:

- **Enabled:** untick to silence the axis.
- **CC, Program change, Velocity or Transpose:** what the axis does. Program change is described in chapter 10, Velocity and Transpose below. The fields that follow are those of the CC mode.
- **CC:** the controller number (0–127).
- **Min and Max:** the value sent at each end of the travel. A Min above Max reverses the direction.
- **Invert:** flips the travel, for example so that pushing a stick up raises the value.
- **The bar** under the fields shows the value being sent right now.

A stick sends its minimum at one edge, its maximum at the other, and the middle of the range at rest.



The stick, trigger and touchpad windows also hold the settings of their loop (Playback, loop CCs, Clear the loop), described in chapter 9.

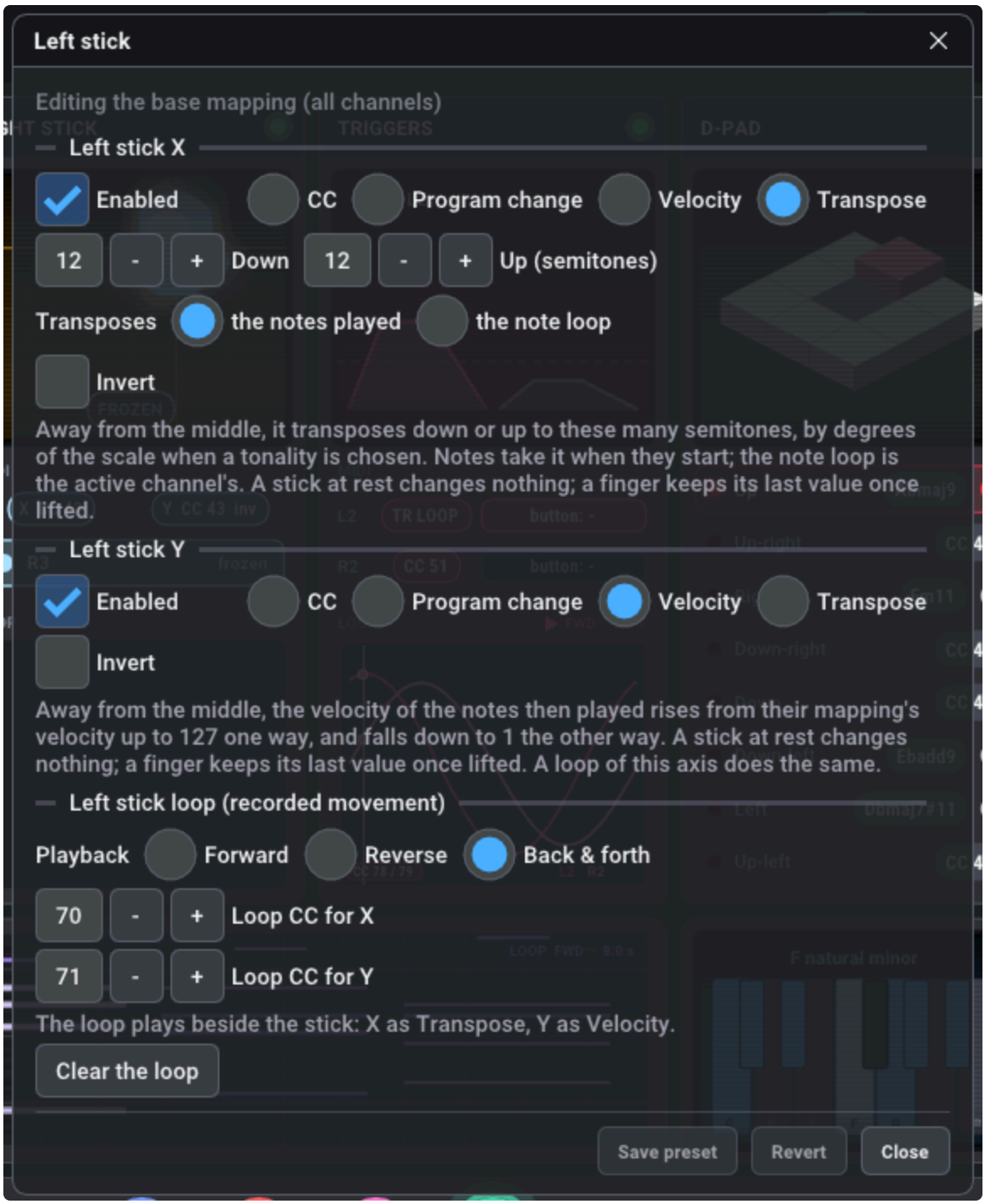

Touchpad
Editing the base mapping (all channels)
Finger 1
X/Y position -> CC
Mode
X (left -> right)
Enabled
CC
Program change
Velocity
Transpose
60
CC
0
Min
127
Max
Invert
61
Y (bottom -> top)
Enabled
CC
Program change
Velocity
Transpose
61
CC
0
Min
127
Max
Invert
68
Finger 1 loop (recorded movement)
Playback
Forward
Reverse
Back & forth
74
Loop CC for X
75
Loop CC for Y
The loop plays beside the finger: X on CC 74, Y on CC 75.
Clear the loop
Finger 2
Save preset
Revert
Close

In the touchpad window, choose each finger's **Mode** first: *Off*, *X/Y position → CC* (two axes, as above) or *4 zones → pads* (a button action for each quarter: top left, top right, bottom left, bottom right). The window scrolls to show finger 2.

> **Note:** when a loop uses the same CC as a live axis, the window shows a red warning: both would send to the same parameter and fight. Choose different CCs.

Velocity and Transpose axes

An axis does not have to send a CC: set it to **Velocity** or **Transpose** and it shapes the notes you play instead. Choose the mode with the radio buttons at the top of the axis form.



Velocity

The axis changes the velocity of the notes then played, starting from the velocity set in their mapping:

- **Stick or finger:** away from the middle one way, the velocity rises towards 127 in proportion to the room left (halfway, a note mapped at 100 plays at 114); the other way, it falls towards 1.

- **Trigger:** pressing it only raises the velocity.
- It applies to the notes played live, not to the note loop.

Transpose

The axis transposes the notes then played, or the note loop:

- **Down** and **Up** set how far it goes at each end of its travel, from 0 to 24 semitones each way. A trigger has **Up** only.
- **Transposes the notes played / the note loop** chooses what it moves. The note loop is the active channel's.
- With a tonality chosen (chapter 7), the axis moves **by degrees of the scale**: 12 semitones make one octave of degrees, notes stay in the key and chords stay chords of the key (in F minor, one degree up turns Fm into Gdim). Without a tonality, it moves by semitones. A note outside the scale first goes to the nearest note of the scale.

What both have in common

- A note takes the value when it starts, and always stops at the pitch it started at.
- A stick at rest changes nothing: it has a small dead zone in the middle.
- A finger on the touchpad keeps its last value once lifted: set a velocity or a transposition, take your finger off, and it stays.
- A frozen axis (Axis freeze) keeps the value it had when it froze.
- A loop of such an axis plays the velocity or transposition instead of CCs: record a stick moving and the transposition follows the recorded gesture.
- Several axes set to the same mode add up.
- The axis pills show the live value: *VEL +40%*, *TR +3*, or *TR LOOP* for an axis set on the note loop.

> **Note:** in headless mode there is no tonality, so Transpose always moves by semitones.

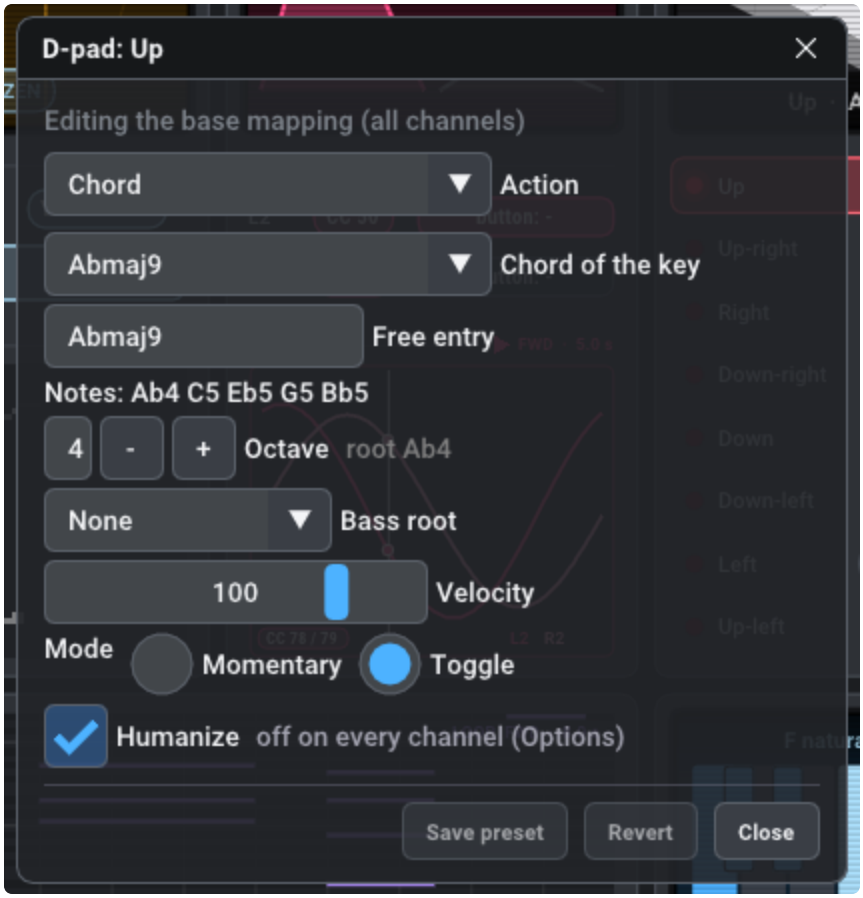
7. Notes, chords and scales

MidiPad plays named chords such as *Fm9* or *Dbmaj7#11*, and can restrict every note and chord menu to the notes of a chosen scale, so you build a playable layout without leaving the key.

Notes

A **Note** action plays one MIDI note. Without a tonality, type its number (0–127); MidiPad shows its name beside it (60 = C4). With a tonality, the field becomes a menu of the notes of the scale on every octave, and the arrows step to the previous or next note of the scale.

Chords



A **Chord** action plays several notes at once, all with the same velocity:

- **Chord:** type a chord name in **Free entry**, or pick one in **Chord of the key** when a tonality is set.
- **Notes:** the notes the chord will play, spelled out, so you can check it before playing it.
- **Octave:** the octave of the root (4 by default: C4 = 60). The other notes stack above it.
- **Bass root:** *None*, *1 octave below* or *2 octaves below*. Doubles the root lower down to anchor the bass, as on the Cross button of the demo, where Fm9 is played over a low F.

A chord whose notes would fall outside the MIDI range (0–127) is refused; lower the octave.

Writing chord names

A chord name is a root (C, C#, Db … B) followed by a type. MidiPad recognises these types:

Family	Types
Triads	(major), m, 5, dim, aug, sus2, sus4
Sixths	6, m6, 69, m69
Sevenths	7, maj7, m7, mmaj7, dim7, m7b5, 7sus4, 7#5, 7b5, maj7#5
Ninths	add9, 9, maj9, m9, mmaj9, 9sus4, m9b5, 7#9, 7b9
Elevenths	11, maj11, m11, m11b5, maj7#11, 9#11
Thirteenths	9b13, 13, maj13, m13

Useful details:

- A capital **M** means major: *CM7* is Cmaj7, not Cm7.
- *C6/9* is accepted for C69.
- A slash chord such as *Ab/C* plays the bass note one octave below the root's octave.

Tonality

Click the **SCALE** block to choose a tonality: a root on the left, a scale on the right. The tonality is shared by all channels and saved with your settings, not with the preset.



Available scales: major, natural minor, harmonic minor, melodic minor, dorian, phrygian, lydian, mixolydian, locrian, major pentatonic, minor pentatonic, blues and whole tone. Choose **None** as the root to remove the tonality.

With a tonality set:

- **Note** menus list only the notes of the scale.
- **Chord of the key** lists, degree by degree, every chord whose notes all belong to the scale.
- **Free entry** stays available for anything else. A note outside the scale is marked *out of key*; a chord with such notes shows *Contains notes outside* followed by the tonality. Both still play. An unknown chord name shows *Unknown chord*, and the previous chord stays active.
- The SCALE block shows the name of the tonality and lights its notes on the keyboard.

> **Tip:** set the tonality before building a preset. The Chord of the key menu then offers only chords that fit, which is the fastest way to fill the D-pad and the face buttons.

8. Channels

MidiPad sends on one MIDI channel at a time, the active channel, and you can switch channels while playing. Point each channel to a different instrument in your DAW and one controller plays a whole band: pads on channel 1, bass on channel 2, drums on channel 16.

Switching channels

- **In the window:** click the arrows of the channel selector in the top bar.
- **From the controller:** give a button the **Channel change** action, Previous (-1) or Next (+1). In the demo, Create steps down and Options steps up. Channels wrap around from 16 to 1.

The CHANNEL block shows the active channel in large digits. Its grid of sixteen channels marks with a violet dot every channel where a loop is playing.

Base mapping and channel mappings

A preset holds a **base mapping**, used by every channel, and optionally a **channel mapping** for any of the sixteen channels. A channel mapping replaces the base mapping entirely while that channel is active: when you switch to it, every control takes its new meaning at once.

The demo preset uses this to give each channel its own role:

Channel	Mapping	Role in the demo
1	Base	Chords on the face buttons and D-pad, CCs on sticks and touchpad
2	Own	Other chords, notes on L2/R2, transport on L1
3–15	Base	Same as channel 1
16	Own	Single notes, touchpad split into four drum pads

The mapping windows always edit what plays now: the mapping of the active channel if it has one, otherwise the base mapping. The scope line at the top of each window tells you which one you are editing.

> **Note:** channel mappings are created in the preset file (chapter 12, key `channels`). Once a channel has its own mapping, you edit it in the window like the base mapping.

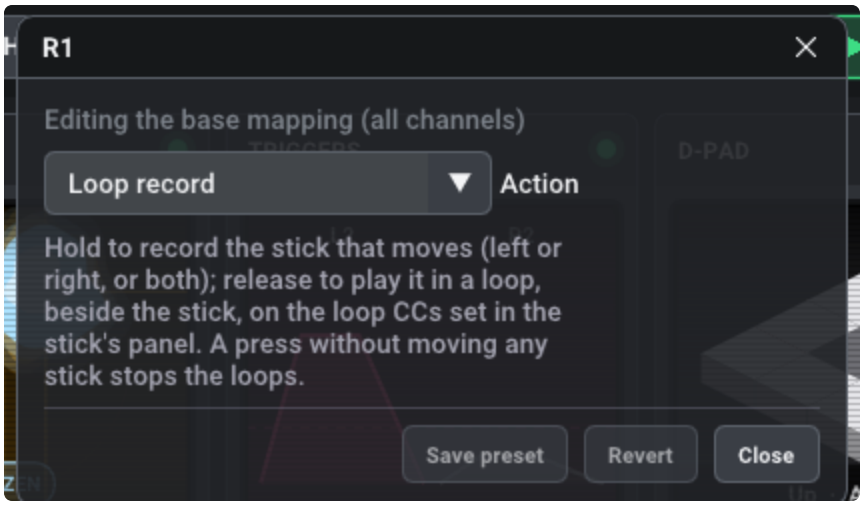
Notes never get stuck

A note always stops on the channel where it started. You can switch channels, or even presets, while holding a chord: its release still reaches the right instrument.

9. Loops and the piano roll

MidiPad records what you play and plays it back in a loop, on the channel where you played it, while your hands stay free for something else. It records two kinds of loops: **movement loops** (sticks, triggers, touchpad fingers), which work like hand-drawn LFOs, and **note loops**, which repeat the notes and chords you played.

Everything starts from a button set to **Loop record** (R1, labelled *Rec*, in the demo).



Recording a movement loop

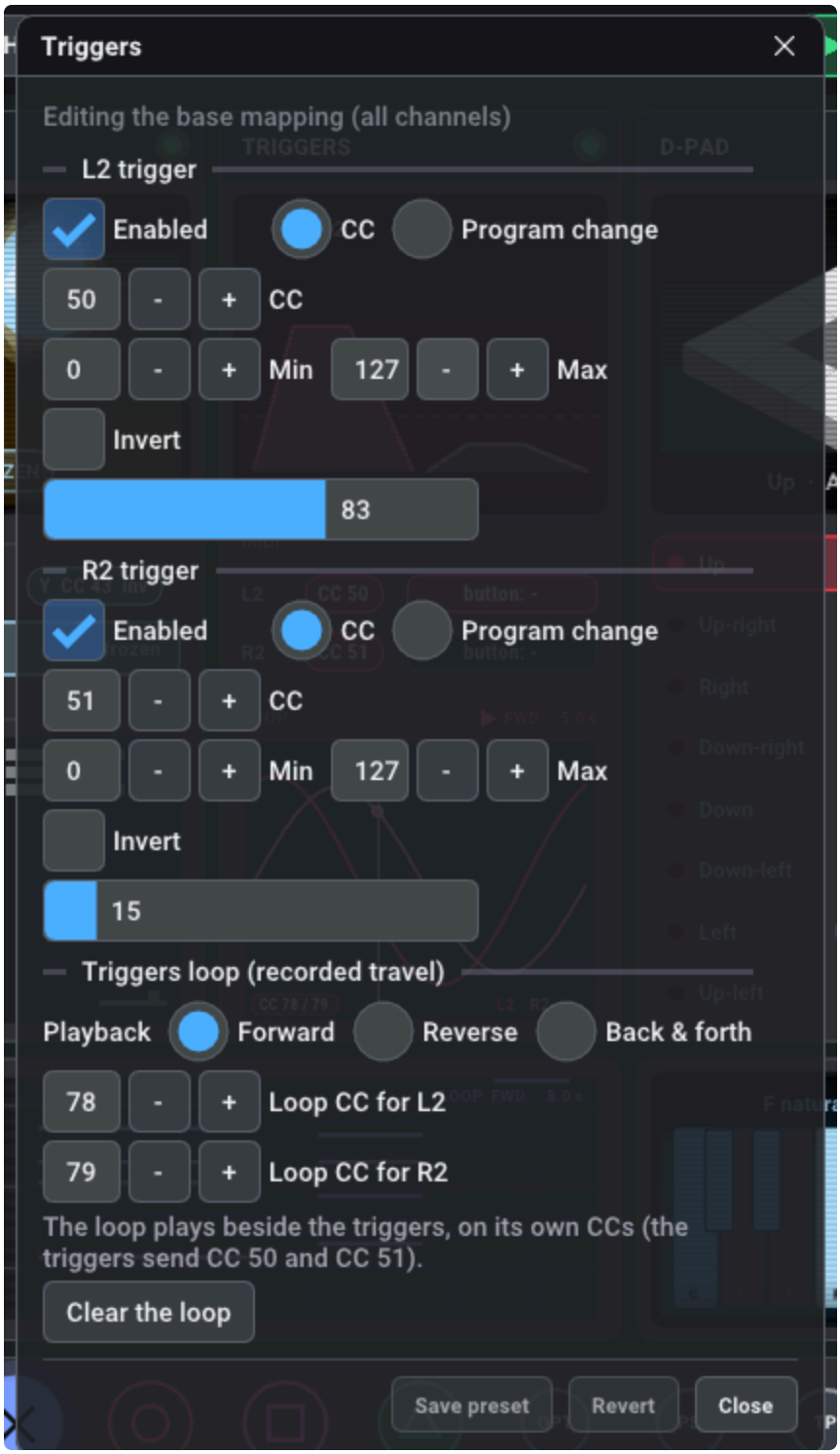
1. Hold **Loop record**.
2. Move a stick, press a trigger or slide a finger on the touchpad. You can move several at once.
3. Release **Loop record**. Each movement starts looping at once, from the start.

Only what moves is recorded: a stick must travel about 12 % from the centre to count. For the touchpad, the moments when the finger was lifted are recorded too, and play nothing. A loop lasts up to 60 seconds.

The loop plays **beside** the control, on CCs of its own, so the stick or finger stays free to play live on its usual CCs. The loop CCs are set in the window of the stick, triggers or touchpad:

Source	Loop CCs (default)
Left stick	70 / 71
Right stick	72 / 73

Source	Loop CCs (default)
Touchpad finger 1	74 / 75
Touchpad finger 2	76 / 77
Triggers L2 / R2	78 / 79



In the same window, **Playback** chooses the direction: *Forward*, *Reverse* or *Back & forth*. **Clear the loop** erases it.

Reading the LOOP panels

The LOOP panel of each module draws the recorded curves of the active channel, with a moving playhead. Its header shows the playback mode and the length (for example *B&F · 4.0 s*, or *1.5 BARS* for a loop synced to Ableton Link), and a pill shows the loop CCs (*CC 70 / 71*). Click the panel to open its settings.

An empty LOOP panel plays a small grey pixel animation, in the style of 1980s video games, so you can tell at a glance which loops are free. It is decoration only.

Recording a note loop

1. Hold **Loop record**.
2. Play notes or chords: face buttons, D-pad, triggers used as buttons, touchpad pads.
3. Release **Loop record**. The notes loop on the channel where you played them, with their velocity and humanization.

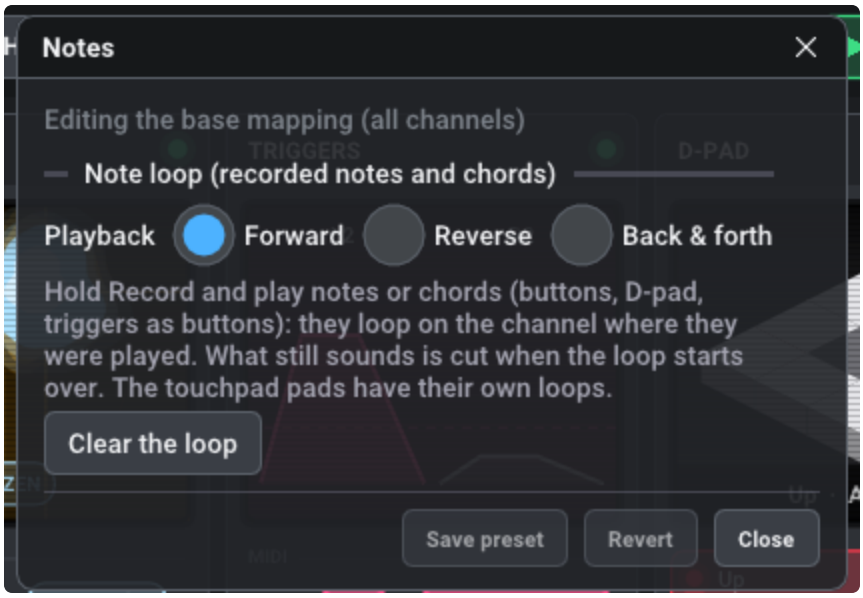
Each channel has one note loop. Recording a new one replaces it. A note still held at the end of the recording (a latched chord) sounds until the end of the loop, and everything sounding stops when the loop starts over.

The piano roll

Without a note loop, the piano roll scrolls the last seconds of the active channel towards the *now* line on the right edge: white for what you play, violet for what a loop plays. Its keyboard follows the range of the notes played.

While a note loop plays, the piano roll becomes a **clip**: the whole loop, from *start* to its end, standing still, with a playhead sweeping it in the playback direction. Notes light up as the playhead crosses them. Beneath the clip, the notes you play live keep scrolling from the right edge in faint cyan, as without a loop. Click the piano roll to choose the playback mode or clear the note loop.





Overdub

Overdub adds notes to a note loop without changing its length.

1. While a note loop plays, tap **Loop record** briefly (less than half a second) without playing anything: overdub is on.
2. Play: every note is added to the loop where it plays.
3. Tap **Loop record** again to turn overdub off.

During overdub, the piano roll shows **OVERDUB** and an orange playhead, each added note appears in orange while you hold it, and the Rec pad is circled in orange.



Stopping and restarting loops

You do	Result
Tap Loop record, no note loop playing, nothing moved	Stops the loops of the active channel
Hold Loop record without playing or moving	Stops the loops of the active channel
Click ■ (Stop) in the top bar	Pauses the loops of every channel, without erasing them
Click ▶ (Play)	Restarts every loop from its start; Play turns green while loops run
Record while loops are paused	Restarts all loops together with the new one

You do	Result
Click PANIC	Stops every loop on every channel and silences everything

Stretching loops in time and amplitude

A movement loop (sticks, triggers, touchpad in axis mode) can be reshaped while it plays, without recording it again, right from the gamepad: stretched or compressed **in time**, and narrowed **in amplitude**. The note loop is not concerned.

Loop stretch: hold a button, move an axis

Give a button or a D-pad direction the **Loop stretch** action. Hold it, then move the stick or the finger whose loop you want to reshape:

- **Stick left or right** compresses or stretches its loop **in time**: up to **100 times longer** (*+100x*) to the right, up to **20 times shorter** (*-20x*) to the left, by steps of a half (*+1.5x*, *+2x*, *+2.5x* …).
- **Stick up or down** widens or narrows its **amplitude** around its middle (halfway between its lowest and highest values): from **0 %** (still) through **100 %** (as recorded) up to **800 %**, by steps of 5 % below 100 % and of 25 % above.
- Widened past 100 %, a swing would run beyond the ends of the CC range (0 and 127), so MidiPad **clips it softly**: the curve bends towards the end, then stays flat there, like an LFO driven into saturation. The bend grows with the amplitude, from none at 100 % to a tenth of the range from 200 %. A small swing widened 3 or 4 times fills the whole range; a wide one widened far turns into a square wave.
- The stick works as a wheel kept turning: nothing in the dead zone around the centre, then faster the further you push. Only the way it mostly moves counts, so a diagonal never changes both at once.
- **On the touchpad** (axis mode), slide a finger: each short slide counts one step for the finger loops, sideways for the length, up and down for the amplitude.
- Crossing from compressed to stretched, the length waits a moment at **1x** (and the amplitude at 100 %, crossing it either way), so the recorded loop is easy to find again.
- **Release the button: everything you set stays.** Recording the loop again puts it back to 1x and 100 %.

Image — The Loop stretch action

While Loop stretch is held, the sticks and fingers only reshape loops: they send no CC and start no recording, and their screens show a **STRETCH** badge with the current factor.

The loop goes on from the same point when it changes, so it never jumps. Stretched, it is smoothed between its recorded points; compressed, each value is the average of the points it passes over; a change of amplitude is eased in over a few milliseconds. A loop of a Velocity or Transpose axis is reshaped the same way.

What you see

While the length or the amplitude changes, a gauge appears at the bottom of the LOOP box: a log scale from -20x to +100x for the length, a scale from 0 % to 800 % for the amplitude, with 100 % marked. It fades out a moment after the last change.

Image — Amplitude gauges while the swing changes: left stick at 60 % (also stretched +2…

Once set, chips stay in the header (*+2.5x*, *AMP 60%*), the header keeps only the length (the small arrows still tell the playback), and the curve is drawn as it plays, narrowed or widened (flat where it is clipped), the recorded one faintly behind. The length shown is the stretched one (*10.0 s*, or *12:30 min* beyond a minute; in beats or bars for a loop synced to Ableton Link).

Image — After: chips in the headers; curves narrowed, or widened and clipped flat at th…

At the computer

The same changes are also at hand with the mouse, over a LOOP panel: **Shift + wheel** for the length, **Ctrl + wheel** for the amplitude (up stretches or widens). Turned quickly, the wheel takes bigger steps.

Playing in time with Ableton Link

By default MidiPad keeps its own time: a loop lasts exactly as long as you held the button. Turn on **Ableton Link** and MidiPad joins the tempo and beat of your DAW, or of any app using Link on the same network: loops then come out a whole number of beats long, start

on the beat and follow the tempo.

To turn it on, click **LINK** in the top bar (it reads *FREE* while Link is off), or choose ☰ › **Options** › SYNC › Ableton Link: **On**. The button turns green and shows the tempo, the number of peers (other Link apps in the session) and a square for each beat of the bar, the current one lit. Alone on the network, MidiPad starts a session of its own at 120 BPM.

Image — Link on at 90 BPM with two peers: the loops measured in beats and bars, over th…

A loop recorded while Link is on:

- **Snaps to whole beats.** When you release Loop record, the length is rounded to the nearest whole number of beats (or bars, see SYNC in Options), one at least. Hold for about six beats and you get exactly six: *1.5 BARS*. One beat, three beats, one bar, two and a half bars: anything goes.
- **Starts on the beat.** Its start moves to the nearest beat, so it stays in phase with the session and with the other loops.
- **Follows the tempo.** It plays in beats, not in seconds: change the tempo in your DAW and the loop speeds up or slows down with it, still in phase.
- **Shows beats and bars.** Its header gives the length in beats or bars (*1 BEAT*, *3 BEATS*, *1 BAR*, *1.5 BARS*), and a grid marks the beats, stronger on the bars, with the bar numbers on top: in the LOOP panels and in the piano roll clip.
- **Still stretches.** Loop stretch works as before: *+2x* makes a one-bar loop last two bars, *-2x* half a bar. Forward, Reverse and B&F all work; in B&F each pass lasts the loop's length.

Note loops and movement loops behave alike.

Start and stop. With *Start / stop sync* on (the default), ▶ and ■ also start and stop the Link session, and MidiPad's loops pause and resume when another app starts or stops it. Turn it off and ▶ / ■ only act on MidiPad.

Free mode. With Link off, nothing changes from earlier versions: loops keep the length you played, to the millisecond. A loop recorded free stays free when you turn Link on; a synced loop goes on at its own pace when you turn Link off.

> **Tip:** Link works over the local network. If MidiPad sees 0 peers while another app has Link on, check that both are on the same network and that no firewall blocks UDP multicast (Link uses port 20808).

Loops on several channels

Each channel has its own loops. Record on channel 1, switch to channel 2, record again: all loops play together, each on its own channel. The LOOP panels and the piano roll always show the loops of the active channel; the CHANNEL block marks the others with a violet dot.

> **Note:** loops live in memory only. They are not saved with the preset and disappear when you quit MidiPad.

10. Program Change and transport

MidiPad can change the sound of your synthesizer with MIDI Program Change messages, from a button or by sweeping an axis, and can send MIDI transport commands.

Program Change from a button

Image — Program change on L1, with Bank select

Give a button the **Program change** action, then choose:

- **Program:** always selects the same program (1–128).
- **Next (+1)** or **Previous (-1):** steps from the last program sent on the channel. The first press sends program 1. With **Wrap around**, stepping past 128 returns to 1 and the other way round; without it, stepping stops at the ends.
- **Bank select** (optional): sends a bank before the program, with **MSB (CC 0)** and/or **LSB (CC 32)**, each switched on separately.

The message goes out on the active channel. The pad label shows *PC 12* for a fixed program, *PC +1* or *PC -1* for a step.

Program Change from an axis

A stick axis, a trigger or a touchpad finger axis can select a program instead of sending a CC. In its axis window, choose **Program change** instead of CC, then set:

- **From** and **To:** the range of programs the travel covers, for example 1 to 16.
- **Invert:** reverses the direction.

- **Bank select:** a bank of its own for this axis, sent before every program.

Aim, wait, release. The program is sent only once the axis has stayed on it for 150 ms, so sweeping across the range does not load every program on the way. Nothing is sent when the axis springs back to rest, so the last program you chose stays. The axis pill shows *PC 1-16*.

Because each axis has its own bank, two axes can browse two different banks: the left stick one set of sounds, the right stick another.

Numbering

MidiPad shows programs from 1 to 128, as most synthesizers do; *Program 1* sends the value 0. Bank MSB and LSB are shown and sent as raw values (0–127). If your synthesizer counts banks from 1, enter the bank number minus 1.

> **Example: Arturia instruments.** Arturia's software instruments select presets inside a playlist: Bank MSB selects the playlist, Bank LSB the song, and the program the preset in that song, all counted from 0. For playlist 1, song 1, set MSB 0 and LSB 0; then *Program 1* loads the first preset of the song.

> **Note:** your DAW must pass Program Change messages on to the plugin. Some hosts do not forward them to VST3 plugins; if nothing happens, check with a MIDI monitor placed before the plugin, and try the VST2 version of the plugin.

Transport

The **Transport** action sends MIDI **Start**, **Stop** or **Continue**. These are system real-time messages: your DAW reacts to them only when it is set to follow external MIDI sync. Many DAWs offer MIDI learn on their transport buttons instead; in that case, use a **CC** action on the button and learn it in the DAW.

11. Options

The Options window holds the settings that belong to your computer rather than to a preset: the MIDI output, the VirMIDI ports, the sync with Ableton Link, the D-pad delay, humanization and the presets folder. Open it from ☰ › **Options**. Settings are saved at once, in `~/.local/share/MidiPad/MidiPad/settings.json`.

Image — The Options window

MIDI INTERFACE

The drop-down menu shows where MidiPad sends its MIDI. Click it to see every output:

Image — Choosing the MIDI output

- **Virtual port "MidiPad"** (*software port*): created by MidiPad itself, for programs that read ALSA sequencer ports.
- **VirMIDI N-M** (*seen as Virtual Raw MIDI …*): kernel MIDI ports. The hint on the right is the name under which programs that only list hardware devices show this port.
- **Hardware interfaces**, such as a USB MIDI interface, to drive an external synthesizer.

Refresh updates the list after you plug in a device. If the saved output is missing at startup, the menu shows it in red, marked *(not found)*, and MidiPad sends to its virtual port until you choose another.

> **Note:** the card number N of a VirMIDI port can change between reboots, depending on the order in which devices are found. MidiPad finds its port again regardless.

VirMIDI ports

Some programs only see the kernel's raw MIDI devices and ignore software ports such as *MidiPad*. The `snd-virmidi` kernel module provides such devices, and MidiPad can load it for you:

1. Under **Ports**, choose how many ports you want, from 1 to 4.
2. Tick **At every startup** to create them again after each reboot.
3. Click **Apply**. Your system asks for the administrator password, in its own dialog.

The status line shows how many ports exist and whether they come back at startup. **Remove** unloads the module and deletes the startup files; close the programs using the ports first.

With *At every startup*, MidiPad writes two small files: `/etc/modules-load.d/midipad-virmidi.conf` and `/etc/modprobe.d/midipad-virmidi.conf`. To load the module by hand instead:

```
sudo modprobe snd-virmidi midi_devs=4
```

Then choose **VirMIDI N-0** in MidiPad and, in the receiving program, the input *Virtual Raw MIDI* (VirMIDI N-1 appears as *Virtual Raw MIDI 2*, and so on).

SYNC

How MidiPad keeps time: on its own (free), or with the tempo and beat of an Ableton Link session. See *Playing in time with Ableton Link* in the Loops chapter.

Image — The SYNC card with Link on

- **Ableton Link**: *Off* (free, the default) or *On*, like the LINK button of the top bar. With Link on, the number of peers and the session's tempo show beside it.
- **Loops snap to**: the unit a loop recorded with Link is rounded to: *1 beat* (the default), *1 bar*, *2 bars* or *4 bars*. *Free* keeps loops free even with Link on.
- **Beats per bar**: 1 to 16 (4 by default), for the bars shown in the headers, the grid and the LINK button.
- **Start / stop sync**: *On* (the default): ▶ and ■ start and stop the Link session, and the loops follow when another app does. *Off*: they only act on MidiPad.

D-PAD

Delay on press and **Delay on release** (0–80 ms, 15 ms by default) set how long a new D-pad direction must hold before it plays. They keep the arrow crossed while pressing or releasing a diagonal from sounding. Set 0 to play every direction at once.

HUMANIZE

Humanization varies velocity and timing at random, for a less mechanical feel.

- **Channels:** the channels it applies to. **All** and **None** select every channel or none.
- **Velocity:** a range in percent of the velocity set in the mapping (100 % by default).
- **Delay:** a range of delay, from 0 to 500 ms.

Every note, including each note of a chord, draws its own velocity and delay within these ranges. Its release is delayed by the same amount, so its length does not change. A control can opt out with the **Humanize** box of its mapping window. PANIC also drops notes still waiting to play.

PRESETS

Folder shows the folder MidiPad reads presets from; **Change…** picks another one. The default is `~/.local/share/MidiPad/MidiPad/presets`.

12. Presets

A preset is a complete layout: the base mapping, the channel mappings and the loop settings, stored in one JSON file. Switch presets to move from one song or setup to another.

The PRESET menu

Image — The PRESET menu

- Click the preset name to open the list: every preset of the presets folder, then **New…**, **Save as…** and **Delete…**.
- The ◀ and ▶ arrows on each side load the previous or next preset.
- **New…** creates an empty preset, asking for its name.
- **Save as…** saves a copy of the current preset under a new name and switches to it.
- **Delete…** deletes the current preset after a confirmation. The file is deleted for good.

Image — Save as

Image — Delete preset

Saving

Every change is heard at once but kept in memory until you save. Press `Ctrl+S` or click **Save preset** in any mapping window. If you open another preset or quit with unsaved changes, MidiPad asks whether to **Save**, **Don't save** or **Cancel**.

MidiPad reopens the last preset used at the next start. To start from a given file instead, pass it on the command line; its folder becomes the presets folder:

```
midipad ~/Music/MidiPad/live-set.json
```

The preset file

Presets are plain text files, `<name>.json`, in the presets folder. You can read, copy, share and edit them in any text editor; after editing a preset that is open in MidiPad, load another preset and come back to it with the arrows to load your changes, rather than clicking **Revert**, which only discards changes made in the window. Here is a short preset: Cross plays a chord, the left stick sends two CCs, and channel 16 has its own mapping.

```
{
    "format": "midipad-preset",
    "version": 2,
    "name": "My Preset",
    "base": {
        "buttons": {
            "south": { "type": "chord", "chord": "Fm9", "octave": 4,
"root_below": 2 },
            "right_shoulder": { "type": "loop_record" }
        },
        "dpad": {
            "up": { "type": "chord", "chord": "Abmaj9", "mode": "toggle" }
        },
        "axes": {
            "left_x": { "cc": 40 },
            "left_y": { "cc": 41, "invert": true }
        }
    },
    "channels": {
        "16": {
```

```
            "buttons": { "south": { "type": "note", "note": 53 } }
        }
    }
}
```

The top level holds `base` (the base mapping) and, optionally, `channels`: one mapping per channel number, from "1" to "16". Each mapping has up to five sections: `buttons`, `dpad`, `axes`, `touchpad` and `loops`. Fields left out take their default value. Chapter 16 lists every key.

> **Note:** if a preset file contains an error, MidiPad refuses it and names the faulty entry, for example `base.buttons.south.note: expected a MIDI note number or a name like "C4"`.

13. Using MidiPad with a DAW on Linux

MidiPad works with any DAW or synthesizer that receives MIDI. The only decision is which port to use, and it depends on how your DAW lists MIDI inputs.

Choosing the port

Your DAW lists…	Output to choose in MidiPad	Input to choose in the DAW
ALSA sequencer ports (most Linux DAWs)	Virtual port "MidiPad"	*MidiPad*
Only hardware MIDI devices	VirMIDI N-0	*Virtual Raw MIDI* (or the name shown next to the port in MidiPad)
JACK MIDI ports only	Virtual port "MidiPad", bridged with `a2jmidid`	*MidiPad* under the a2j bridge
An external synthesizer	Your USB MIDI interface	—

When in doubt, use a VirMIDI port: every program sees it.

Assigning CCs with MIDI learn

Most DAWs and synthesizers assign a CC to a parameter by MIDI learn: you click the parameter, then move the control. Sticks send two CCs at once, which can confuse MIDI learn. To learn one axis alone:

1. Open the stick window and untick **Enabled** for the axis you do not want to learn.
2. Start MIDI learn in the DAW and move the stick.
3. Tick **Enabled** again.

The same applies to loops: their CCs (70–79 by default) are separate from the live CCs, so learn them separately. Record a short loop, then start MIDI learn while it plays.

> **Tip:** use **Axis freeze** to hold a value steady while you assign or check a parameter.

A multitimbral setup

1. Create one instrument track per sound, each listening to the MidiPad port on its own MIDI channel.
2. In MidiPad, give each channel the layout it needs (chapter 8).
3. Put **Channel change** on two buttons to move between instruments while playing.
4. Record loops on each channel: they keep playing while you move on to the next instrument.

Latency

MidiPad reads the controller 1,000 times per second on a dedicated thread, with real-time priority when the system allows it, and sends MIDI directly from that thread. Drawing the window never delays the MIDI. On a typical machine, processing takes about 20 µs per reading. For the lowest latency, prefer a USB connection to Bluetooth.

> **Note:** for real-time priority, your user must be allowed to use it, usually by being in the `audio` group with the limits most audio distributions set up.

14. Running without a window

MidiPad can play a preset without its window, from a terminal: useful on a small computer dedicated to the stage, or to start MidiPad from a script. It reads the first controller connected, before or after the start, and creates a virtual MIDI port named *MidiPad* unless

you choose another output.

```
midipad --headless ~/.local/share/MidiPad/MidiPad/presets/"Electronica
Demo.json"
```

The preset reloads automatically whenever its file is saved, so you can edit it in a text editor and hear the result at once. If the new version contains an error, the previous one stays active and the error is printed.

Option	Effect
`-c`, `--connect NAME`	Send to an existing port whose name contains NAME, instead of creating a virtual port (example: `-c "VirMIDI 0-0"`)
`-n`, `--port-name NAME`	Name of the virtual port (default: MidiPad)
`--api alsa|jack`	MIDI system to use on Linux (default: alsa)
`--channel N`	Starting MIDI channel, 1 to 16 (default: 1)
`-m`, `--monitor`	Print every MIDI message sent
`-i`, `--input`	Print the buttons pressed
`--stats`	Print the measured processing times on exit
`--poll-hz N`	Controller readings per second (default: 1000)
`--list-ports`	List the MIDI output ports and exit
`-h`, `--help`	Show the help

Press `Ctrl+C` to stop. Loops and all other features of the preset work as in the window; the Options window's settings do not apply: the D-pad delay stays at 15 ms and humanization is off.

15. Troubleshooting

Most problems come from the route between MidiPad and the receiving program. Check that route first: MidiPad's status bar shows where it sends, and the command `aseqdump` shows what it actually sends.

Checking what MidiPad sends

Switch MidiPad's output to the virtual port "MidiPad" for the test, then watch it while you play:

```
aseqdump -l
```

```
aseqdump -p MidiPad
```

Every press and movement prints a line: *Note on*, *Control change*, *Program change* and so on, with its channel. If lines appear, MidiPad works and the problem lies in the route or the receiving program. The first command lists every port; a VirMIDI port cannot be watched this way, because it shows what leaves the port on the other side, not what MidiPad sends into it.

Common problems

Symptom	Cause	Solution
The status bar shows *No gamepad*	Controller not connected or not paired	Reconnect it by USB, or pair it again over Bluetooth (chapter 2)
+ touchpad is missing, the TOUCHPAD module stays still	No access to the touchpad device	Install the udev rule (chapter 2) and reconnect the controller
The DAW does not list MidiPad	The DAW only lists hardware devices	Use a VirMIDI port (chapter 11)
The MIDI output window appears at every start	The saved port is missing	Choose a port again; with VirMIDI, tick **At every startup**
No sound, but `aseqdump` shows messages	The track does not listen to the port or channel	Set the track's input to the port, on all channels; arm or monitor the track
A stick moves two parameters during MIDI learn	Both axes send at once	Untick **Enabled** for the other axis while learning (chapter 13)

Symptom	Cause	Solution
A parameter jumps between two values	A loop and a live axis use the same CC	Give the loop its own CCs (red warning in the window)
The D-pad plays a wrong direction on diagonals	Delay too short for your playing	Raise the D-pad delay in Options
Program changes have no effect	The host does not pass them to the plugin, or the numbering differs	Check with a MIDI monitor before the plugin; try the VST2 version; check the bank numbering (chapter 10)
Start / Stop do nothing in the DAW	The DAW does not follow external MIDI sync	Enable external sync, or use a CC with MIDI learn (chapter 10)
Notes keep sounding	An instrument missed a note-off	Click **PANIC**
VirMIDI: *The VirMIDI ports are in use*	A program still has the ports open	Close it, then **Apply** or **Remove** again
VirMIDI: *Not authorized*	The password was refused or the dialog closed	Click **Apply** again and enter the administrator password

Reporting a problem

When something behaves unexpectedly, note the MidiPad version, the controller and how it is connected (USB or Bluetooth), the output port, and the output of `aseqdump` while you reproduce the problem.

16. Reference

Keyboard shortcuts

Shortcut	Action
`Ctrl+S`	Save the preset
`Ctrl+Q`	Quit (asks about unsaved changes)

Shortcut	Action
`F11`	Fullscreen on or off
`Esc`	Close the window or dialog in front

Files and folders

Item	Location
Program	`build/release/src/app/midipad` after building
Settings	`~/.local/share/MidiPad/MidiPad/settings.json`
Presets (default)	`~/.local/share/MidiPad/MidiPad/presets/`
Touchpad access rule	`/etc/udev/rules.d/70-dualsense.rules`
VirMIDI at startup	`/etc/modules-load.d/midipad-virmidi.conf`, `/etc/modprobe.d/midipad-virmidi.conf`

Control names

The names used in preset files, with the DualSense labels shown in the window:

Preset key	DualSense	Section
`south`, `east`, `west`, `north`	Cross, Circle, Square, Triangle	`buttons`
`left_shoulder`, `right_shoulder`	L1, R1	`buttons`
`left_trigger`, `right_trigger`	L2, R2 (as buttons)	`buttons`
`left_stick`, `right_stick`	L3, R3 (stick clicks)	`buttons`
`back`, `start`, `guide`	Create, Options, PS	`buttons`
`touchpad`, `misc1`	Touchpad click, Mute	`buttons`
`up`, `up_right`, `right`, `down_right`, `down`, `down_left`, `left`, `up_left`	D-pad directions	`dpad`

Preset key	DualSense	Section
`left_x`, `left_y`, `right_x`, `right_y`	Stick axes	`axes`
`left_trigger`, `right_trigger`	L2, R2 (as axes)	`axes`
`finger1`, `finger2`	Touchpad fingers	`touchpad`
`top_left`, `top_right`, `bottom_left`, `bottom_right`	Touchpad zones	`touchpad.fingerN.zones`
`left`, `right`, `finger1`, `finger2`, `triggers`, `notes`	Loops	`loops`

Action fields

`type`	Fields (default)
`note`	`note` (number or name such as "C4"), `velocity` (100), `mode` ("momentary" or "toggle"), `humanize` (true)
`cc`	`cc`, `value` (127), `off_value` (0), `mode`
`chord`	`chord` (name), `octave` (4), `root_below` (0, 1 or 2), `velocity`, `mode`, `humanize`
`transport`	`command`: "start", "stop" or "continue"
`channel`	`step`: 1 or -1
`freeze`	`axes`: list of axis names
`loop_record`	—
`program`	`program` (0–127) or `step` (1 or -1) with `wrap` (true); `bank_msb`, `bank_lsb` (optional)
`loop_stretch`	—

Axis fields

Field	Meaning (default)
`cc`	Controller number; required unless `program` is true
`min`, `max`	Values at the ends of the travel (0, 127); programs 0–127 in program mode
`invert`	Reverse the travel (false)
`program`	Select programs instead of sending a CC (false)
`bank_msb`, `bank_lsb`	Bank sent before each program, in program mode (none)
`velocity`	true: a Velocity axis instead of a CC (false)
`transpose`	A Transpose axis: {"down": 12, "up": 12, "to": "notes"}, down and up from 0 to 24 semitones, "to": "notes" or "loop"

A touchpad finger takes `mode`: "axis" (with `x` and `y`, each an axis entry) or "zones" (with `zones`, each an action). A loop entry takes `playback` ("forward", "reverse" or "back_and_forth") and `cc_x`, `cc_y`.

Timing values

Value	Setting
Controller readings	1,000 per second
D-pad delay	15 ms on press and release, 0–80 ms in Options
Trigger as button	Press at 25 %, release at 15 % of the travel
Program Change from an axis	Sent after 150 ms on the same program
Overdub tap	Shorter than 500 ms
Loop recording	200 samples per second, 60 s at most
Note loop	Up to 4,096 note events per channel

Value	Setting
Humanize delay	0–500 ms
Loop length (stretch)	-20x to +100x, steps of 0.5; Loop stretch + X (or Shift + wheel)
Loop amplitude	0 % to 800 %, steps of 5 % up to 100 % and of 25 % above; softly clipped at the ends of the CC range; eased in; Loop stretch + Y (or Ctrl + wheel)
Loop stretch with a stick	Dead zone 25 % of the half travel; 2 to 40 length steps, 2 to 20 amplitude steps a second; 0.4 s pause at 1x and at 100 %
Loop stretch with a finger	One step every 3 % of the touchpad's width or height
Loop sync (Ableton Link)	Length rounded to whole snaps (1 beat by default), one at least; start moved to the nearest snap; 4 beats per bar by default (1–16)
Transpose	Up to 24 semitones each way; with a tonality, 12 semitones = one octave of degrees